

Binomial Coefficients

$$\binom{n}{k}$$

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Binomial Coefficients

$\binom{n}{k}$ is the number of ways we can choose k distinct elements from a set of n elements

$$\binom{n}{k} - n \text{ choose } k$$

In other words, $\binom{n}{k}$ is the number of k -element subsets of a set of size n .

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Binomial Theorem

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k}$$

Proof:

$$(a + b)^n = \underbrace{(a + b) \cdots (a + b)}_{n \text{ times}}$$

➤ There are $\binom{n}{k}$ ways to “mark” k brackets as a . Thus, the coefficient for $a^k b^{n-k}$ is $\binom{n}{k}$.

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Binomial Distribution

Let $p + q = 1$. Consider n i.i.d Bernoulli trials

$$X_1, \dots, X_n$$

For each i ,

$$\Pr(X_i = 1) = p; \Pr(X_i = 0) = q$$

Then, the sum $X_1 + \dots + X_n$ has the Binomial distribution with parameters n, p . Specifically:

$$\Pr\left(\sum_i X_i = k\right) = \binom{n}{k} p^k q^{n-k}$$

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Then, the sum $X_1 + \dots + X_n$ has the Binomial distribution with parameters n, p . Specifically:

$$\begin{aligned} \Pr\left(\sum_{i=1}^n X_i = k\right) &= \sum_{|S|=k} \Pr\left(X_i = \begin{cases} 1, & i \in S \\ 0, & i \notin S \end{cases} \forall i\right) \\ &= \binom{n}{k} p^k q^{n-k} \end{aligned}$$

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Formula for Binomials

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Proof: Count # of *ordered* sets of size k : We can choose the 1^{st} element in n ways, 2^{nd} element in $(n-1)$ ways, i^{th} element in $(n+1-i)$ ways.

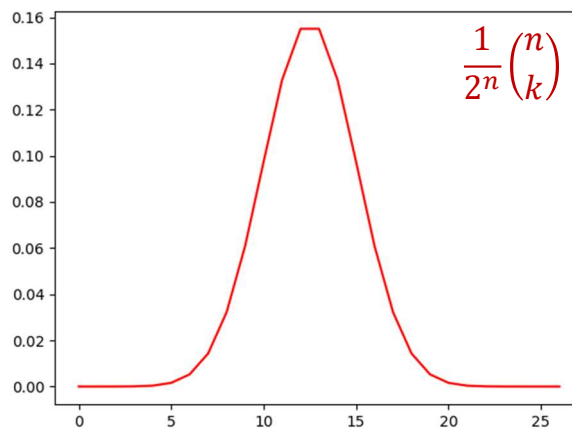
Number of ordered k -subsets:

$$n(n-1)\cdots(n-k+1) = \frac{n!}{(n-k)!}$$

Every *unordered* set is counted $k!$ time.

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Central Limit Theorem

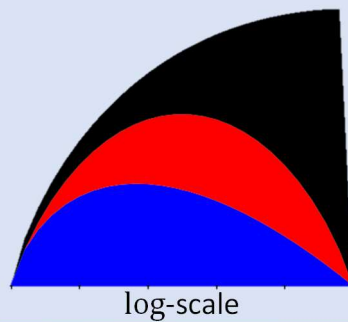


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Useful Inequalities

$$\left(\frac{n}{k}\right)^k \leq \binom{n}{k} < \left(\frac{en}{k}\right)^k$$

These bounds are convenient for estimating various probabilities.



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Useful Inequalities

$$\left(\frac{n}{k}\right)^k \leq \binom{n}{k} < \left(\frac{en}{k}\right)^k$$

- Please, keep in mind that

$$\binom{n}{k} = \binom{n}{n-k}$$

- Make sure that you use this bound with

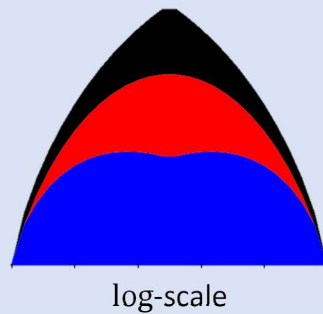
$$k' = \min(k, n - k)$$

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1st Inequality

$$\left(\frac{n}{k}\right)^k \leq \binom{n}{k} < \left(\frac{en}{k}\right)^k$$

These bounds are fairly tight and very useful for estimating various probabilities.

Proof:

$$\binom{n}{k} = \frac{n(n-1)\cdots(n-k+1)}{k(k-1)\cdots 1} = \prod_{i=0}^{k-1} \frac{n-i}{k-i}$$

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1st Inequality

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These bounds are fairly tight and very useful for estimating various probabilities.

Proof:

$$\binom{n}{k} = \prod_{i=0}^{k-1} \frac{n-i}{k-i} \geq \prod_{i=0}^{k-1} \frac{n}{k} = \left(\frac{n}{k}\right)^k$$

Because $(n-i)k \geq (k-i)n$.

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2nd Inequality

$$\left(\frac{n}{k}\right)^k \leq \binom{n}{k} < \left(\frac{en}{k}\right)^k$$

Proof:

$$\binom{n}{k} = \left(\frac{n!}{(n-k)!}\right) \cdot \frac{1}{k!} \leq \frac{n^k}{k!} < \left(\frac{en}{k}\right)^k$$

Because $k! > \left(\frac{k}{e}\right)^k$:

$$e^k = 1 + k + \frac{k^2}{2!} + \cdots + \frac{k^k}{k!} + \cdots$$

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